

Lecture 12: The Second Law of Thermodynamics – Tutorial Sheet

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Use the Master Procedure: identify the system, the reservoirs and the process (heat flow, engine, refrigerator, expansion), sketch (the energy-flow diagram or the pV cycle), convert units (kelvin; magnitudes of Q_H and Q_C), predict (allowed or not, the sign of ΔS , bounds on e or K), select the model, execute, evaluate. Problems A1 and A2 are solved in full; final answers to the others are at the end. Useful values: $R = 8.314 \text{ J/(mol K)}$; $c_{\text{water}} = 4186 \text{ J/(kg K)}$; $L_f = 3.34 \times 10^5$ and $L_v = 2.256 \times 10^6 \text{ J/kg}$ (water); $\gamma = 1.67$ (monatomic), 1.40 (diatomic); $C_V = \frac{3}{2}R$ (monatomic), $\frac{5}{2}R$ (diatomic), $C_p = C_V + R$. Throughout, Q_H and Q_C in the engine formulas are magnitudes: $e = W/Q_H = 1 - Q_C/Q_H$, $K = Q_C/W$, $e_C = 1 - T_C/T_H$, $K_C = T_C/(T_H - T_C)$.

A. Worked examples

A1. Mixing water

0.30 kg of water at 80°C is poured into 0.60 kg of water at 20°C in an insulated container. Find the final temperature and the total entropy change, and decide whether the process is reversible.

Solution. *Predict:* heat flows from hot to cold across a finite temperature difference, so the process is irreversible and $\Delta S_{\text{total}} > 0$. *Final temperature* (nothing else is heated, so $\sum Q = 0$): $m_h(T_h - T_f) = m_c(T_f - T_c)$ gives $T_f = \frac{(0.30)(353.15) + (0.60)(293.15)}{0.90} = 313.15 \text{ K} = 40^\circ\text{C}$. *Entropy* (each portion goes from its own initial temperature to T_f ; calculate along a reversible heating or cooling path): $\Delta S_h = m_h c \ln \frac{T_f}{T_h} = (0.30)(4186) \ln \frac{313.15}{353.15} = -151.0 \text{ J/K}$; $\Delta S_c = (0.60)(4186) \ln \frac{313.15}{293.15} = +165.8 \text{ J/K}$. $\Delta S_{\text{total}} = +14.8 \text{ J/K} > 0$. *Check:* the heat exchanged is $Q = m_h c (T_h - T_f) = (0.30)(4186)(40) = 5.02 \times 10^4 \text{ J}$. The hot water loses it at an average temperature of about 333 K and the cold water gains it at about 303 K, so $\Delta S \approx 5.02 \times 10^4 (1/303 - 1/333) = +14.9 \text{ J/K}$, in agreement. The process is irreversible: a positive total.

A2. A three-step engine

0.100 mol of a monatomic ideal gas at 300 K and volume V_1 is taken through the cycle: (1→2) isobaric expansion to $2V_1$; (2→3) isochoric cooling to 300 K; (3→1) isothermal compression back to V_1 . (a) Sketch the cycle on a pV diagram. (b) Find Q , W for each leg. (c) Find the efficiency and compare it with the Carnot efficiency for the extreme temperatures.

Solution. (a) State 1 is at (V_1, p) ; state 2 at $(2V_1, p)$ with $T_2 = 600 \text{ K}$; state 3 at $(2V_1, p/2)$ with $T_3 = 300 \text{ K}$, on the same isotherm as state 1. The path goes right on the top, down, and back along the isotherm: *clockwise*, so $W_{\text{net}} > 0$ (an engine). *Predict:* the isobaric leg takes heat in; the isochoric leg and the isothermal compression give heat out. (b) $nRT_1 = (0.100)(8.314)(300) = 249.4 \text{ J}$. 1→2: $W = p\Delta V = nR\Delta T = 249.4 \text{ J}$; $Q = nC_p\Delta T = \frac{5}{2}(249.4) = 623.6 \text{ J}$ (in). 2→3: $W = 0$; $Q = nC_V\Delta T = \frac{3}{2}(0.100)(8.314)(-300) = -374.1 \text{ J}$ (out). 3→1: $W = Q = nRT_1 \ln(V_1/2V_1) = -172.9 \text{ J}$ (out). (c) $W_{\text{net}} = 249.4 + 0 - 172.9 = 76.5 \text{ J}$; $Q_H = 623.6 \text{ J}$; $Q_C = 374.1 + 172.9 = 547.0 \text{ J}$. $e = \frac{76.5}{623.6} = 0.123$. *Check:* $Q_H - Q_C = 623.6 - 547.0 = 76.6 \text{ J} = W$. $\checkmark$ $e_C = 1 - 300/600 = 0.500$, so the cycle reaches only a quarter of the Carnot limit: a lot of the heat is rejected on the isochoric leg.

B. Practice

B1. Boiling and the reservoir

0.100 kg of water at 100°C is boiled to steam at 100°C and 1 atm by a heater at 500 K . (a) Find ΔS of the water. (b) Find ΔS of the heater. (c) Find ΔS_{total} . Is the process reversible? (d) What temperature of the heater would make it reversible?

B2. A hot block in a lake

A 2.0 kg copper block ($c = 385\text{ J}/(\text{kg K})$) at 400 K is dropped into a large lake at 290 K . Find the entropy change of the block, of the lake (treat its temperature as constant) and the total.

B3. A conducting rod

A copper rod joins a reservoir at 500 K to one at 300 K . In steady state 3000 J flows through the rod. Find the entropy change of each reservoir and of the whole system, and say whether the rod's entropy changes in steady state.

B4. Free expansion of a gas

2.0 mol of an ideal gas expands freely and adiabatically into a vacuum until its volume has doubled. (a) Find ΔT , Q and W . (b) Find the entropy change of the gas, using a reversible isothermal path. (c) Find ΔS_{total} . (d) Explain, with this result, why the gas never returns by itself to its original volume.

B5. A petrol engine

An engine delivers 40 kW at a thermal efficiency of 0.28 . Find the rate at which heat is supplied, the rate at which heat is rejected, and the rate at which fuel is burned (fuel energy $4.6 \times 10^7\text{ J/kg}$).

B6. A rectangular cycle

A diatomic ideal gas goes round the cycle $1 \rightarrow 2 \rightarrow 3 \rightarrow 4 \rightarrow 1$: 1 ($2.0 \times 10^5\text{ Pa}$, 1.0 L), 2 ($4.0 \times 10^5\text{ Pa}$, 1.0 L), 3 ($4.0 \times 10^5\text{ Pa}$, 3.0 L), 4 ($2.0 \times 10^5\text{ Pa}$, 3.0 L). (a) Sketch the cycle and state its direction. (b) Make a table of W , ΔU and Q for each leg. (c) Find the efficiency. (d) Find the Carnot efficiency for the highest and lowest temperatures reached.

B7. An Otto engine

An Otto engine has $r = 9.0$ and $\gamma = 1.40$. (a) Find e . (b) At the start of compression the air is at 290 K and $1.013 \times 10^5\text{ Pa}$: find T and p after the adiabatic compression. (c) What compression ratio gives $e = 0.60$? (d) A real engine with this r has $e = 0.30$: what fraction of the ideal efficiency is that?

B8. A refrigerator

A refrigerator has $K = 4.0$ and an input power of 120 W . (a) At what rate does it remove heat from the interior? (b) At what rate does it release heat into the kitchen? (c) How long does it take to freeze 0.50 kg of water already at 0°C , if only this heat flow is considered?

B9. The Carnot cycle of a gas

0.100 mol of an ideal gas is the working substance of a Carnot engine between 500 K and 300 K. In the isothermal expansion at 500 K the volume triples. (a) Find Q_H , Q_C , W and e . (b) Find the volume ratio V_c/V_d for the isothermal compression. (c) A different real engine works between reservoirs at 700 K and 300 K, absorbs 5000 J per cycle and achieves half the Carnot efficiency for those reservoirs: find its W , Q_C and ΔS_{total} per cycle, and compare with a Carnot engine absorbing the same heat.

B10. Possible or impossible?

Decide for each, using e_C or K_C and ΔS_{total} : (a) an engine between 400 K and 300 K with $e = 0.30$; (b) a refrigerator between 270 K and 300 K with $K = 12$; (c) an engine that absorbs 2000 J at 500 K and rejects 1200 J at 300 K; (d) an engine that absorbs 1000 J at 600 K and rejects 700 J at 300 K. For the possible ones state whether they are reversible.

B11. Heating a house

A house at 21°C needs 15 kW of heating when the outside air is at 4°C. (a) Find the maximum possible K_{hp} of a heat pump. (b) A real heat pump has $K_{\text{hp}} = 3.5$: find its electrical power. (c) Compare with a resistance heater.

B12. Ocean thermal energy

An ocean thermal plant uses surface water at 27°C as the hot reservoir and deep water at 5°C as the cold reservoir. (a) Find the maximum efficiency. (b) What is the minimum rate of heat input needed for 10 MW of output? (c) Why is the real efficiency lower still?

B13. Challenge: the method is not given

A Carnot engine operating between 600 K and 300 K drives a Carnot refrigerator operating between 300 K and 250 K. For each 1000 J absorbed by the engine from its hot reservoir: (a) find the heat the refrigerator removes from its cold space; (b) find the heat delivered to the 300 K reservoir in total; (c) show that the total entropy change of the four reservoirs is zero, and explain.

Exam-style practice

Ten five-option items in the style of the course examination, **suggested time: 3 minutes per item**. For each item sketch first, predict the sign or size, then calculate (IDENTIFY, SET UP, EXECUTE, EVALUATE). Choose one answer, A to E. Items marked as sharing information are linked. Answers and brief solutions are at the end of this sheet. Take $g = 9.80 \text{ m/s}^2$.

E1. A heat engine absorbs 800 J from a hot reservoir and rejects 520 J to a cold reservoir in each cycle. Its thermal efficiency is

- (A) 0.28 (B) 0.35 (C) 0.54 (D) 0.65 (E) 2.86

Items E2 and E3 share this information: A Carnot engine operates between reservoirs at 500 K and 300 K.

E2. Its efficiency is

- (A) 0.25 (B) 0.60 (C) 0.67 (D) 0.80 (E) 0.40

E3. If the engine absorbs 2000 J per cycle, the work it does per cycle is

- (A) 800 J (B) 500 J (C) 1200 J (D) 1600 J (E) 2000 J

E4. Which of the following statements is/are true?

- I. No cyclic engine can convert all the heat it absorbs from a single reservoir into work (Kelvin statement).
- II. Heat cannot flow spontaneously from a colder body to a hotter body with no other effect (Clausius statement).
- III. The total entropy of an isolated system can decrease in a spontaneous process.

- (A) I only (B) I and II only (C) I and III only (D) II and III only (E) I, II and III

Items E5 and E6 share this information: A refrigerator removes 400 J of heat from the cold space in each cycle using 100 J of work.

E5. Its coefficient of performance is

- (A) 0.25 (B) 0.80 (C) 3.00 (D) 4.00 (E) 5.00

E6. The heat released to the kitchen (the hot reservoir) in each cycle is

- (A) 100 J (B) 300 J (C) 400 J (D) 500 J (E) 800 J

E7. The entropy change of 0.500 kg of ice melting reversibly at 0 °C ($L_f = 3.34 \times 10^5$ J/kg) is

- (A) 306 J/K
(B) 448 J/K
(C) 611 J/K
(D) 1223 J/K
(E) 1.67×10^5 J/K

E8. A 0.50 kg copper block ($c = 390$ J kg⁻¹ K⁻¹) at 360 K is dropped into a large lake at 300 K and cools to 300 K. The total entropy change of block plus lake is

- (A) -35.55 J/K
(B) -3.45 J/K
(C) 0 J/K
(D) +3.45 J/K
(E) +39.00 J/K

E9. An ideal Otto engine has compression ratio 8.0 and $\gamma = 1.40$. Its thermal efficiency is

- (A) 0.33 (B) 0.44 (C) 0.56 (D) 0.75 (E) 0.88

E10. Three inventors each claim an engine operating between reservoirs at 500 K and 300 K. Which claim(s) could be possible without violating the second law?

- I. It absorbs 1000 J and rejects 700 J per cycle.
- II. It absorbs 1000 J and does 450 J of work per cycle.
- III. It absorbs 800 J and rejects 500 J per cycle.

- (A) I only (B) I and II only (C) I and III only (D) II and III only (E) I, II and III

Answers

- B1 (a) $\Delta S = mL_v/T = (0.100)(2.256 \times 10^6)/373.15 = +605 \text{ J/K}$. (b) $-2.256 \times 10^5/500 = -451 \text{ J/K}$. (c) $+153 \text{ J/K} > 0$: irreversible (finite ΔT). (d) 373.15 K (an infinitesimally hotter heater)
- B2 $Q = mc\Delta T = 8.47 \times 10^4 \text{ J}$ to the lake. $\Delta S_{\text{block}} = mc \ln(290/400) = -247.6 \text{ J/K}$; $\Delta S_{\text{lake}} = 8.47 \times 10^4/290 = +292.1 \text{ J/K}$; total $+44.4 \text{ J/K}$
- B3 $\Delta S_{500} = -6.0 \text{ J/K}$; $\Delta S_{300} = +10.0 \text{ J/K}$; total $+4.0 \text{ J/K}$. The rod is in a steady state, so its own entropy does not change: all the entropy is produced in the rod and goes into the cold reservoir
- B4 (a) $\Delta T = 0$ (ideal gas, $\Delta U = 0$), $Q = 0$, $W = 0$. (b) $\Delta S = nR \ln 2 = (2.0)(8.314)(0.693) = +11.5 \text{ J/K}$. (c) The surroundings do not change: $\Delta S_{\text{total}} = +11.5 \text{ J/K}$. (d) A spontaneous return would give $\Delta S_{\text{total}} < 0$; the number of ways falls by a factor of 2^N (about $2^{10^{24}}$)
- B5 $P_H = 40/0.28 = 143 \text{ kW}$; $P_C = 143 - 40 = 103 \text{ kW}$; fuel rate $= 1.43 \times 10^5/4.6 \times 10^7 = 3.1 \times 10^{-3} \text{ kg/s}$ (11.2 kg/h)
- B6 (a) clockwise (right on the top at 4×10^5). (b) pV values 200, 400, 1200, 600 J. 1→2: $W = 0$, $\Delta U = +500$, $Q = +500$; 2→3: $W = +800$, $\Delta U = +2000$, $Q = +2800$; 3→4: $W = 0$, $\Delta U = -1500$, $Q = -1500$; 4→1: $W = -400$, $\Delta U = -1000$, $Q = -1400$ (all in J). Net $W = +400$, $\Delta U = 0$. (c) $Q_H = 3300 \text{ J}$; $e = 400/3300 = 0.121$. (d) $T_{\text{max}}/T_{\text{min}} = 1200/200 = 6$; $e_C = 1 - 1/6 = 0.833$
- B7 (a) $e = 1 - 9.0^{-0.40} = 0.585$. (b) $T = 290(9.0)^{0.40} = 698 \text{ K}$; $p = p_1 r^\gamma = (1.013 \times 10^5)(9.0)^{1.40} = 2.20 \times 10^6 \text{ Pa}$ (about 22 atm). (c) $r^{0.40} = 2.5$, $r = 2.5^{2.5} = 9.9$. (d) $0.30/0.585 = 0.51$
- B8 (a) $Q_C/t = KP = 480 \text{ W}$. (b) $480 + 120 = 600 \text{ W}$. (c) $Q = mL_f = 1.67 \times 10^5 \text{ J}$; $t = 1.67 \times 10^5/480 = 348 \text{ s}$ (about 5.8 min)
- B9 (a) $Q_H = nRT_H \ln 3 = (0.100)(8.314)(500)(1.0986) = 457 \text{ J}$; $Q_C = nRT_C \ln 3 = 274 \text{ J}$; $W = 183 \text{ J}$; $e = 0.40 = 1 - 300/500$. (b) $V_c/V_d = V_b/V_a = 3$. (c) $e_C = 1 - 300/700 = 0.571$, so the real engine has $e = 0.286$: $W = 1429 \text{ J}$, $Q_C = 3571 \text{ J}$, $\Delta S_{\text{total}} = 3571/300 - 5000/700 = +4.76 \text{ J/K}$; the Carnot engine would give $W = 2857 \text{ J}$, $Q_C = 2143 \text{ J}$, $\Delta S_{\text{total}} = 0$. The lost work $1429 \text{ J} = T_C \Delta S_{\text{total}} = (300)(4.76)$
- B10 (a) $e_C = 0.25$: impossible. (b) $K_C = 270/30 = 9$: impossible. (c) $e = 0.40 = e_C$, $\Delta S_{\text{total}} = 1200/300 - 2000/500 = 0$: possible and reversible. (d) $e = 0.30 < e_C = 0.50$, $\Delta S_{\text{total}} = 700/300 - 1000/600 = +0.67 \text{ J/K}$: possible, irreversible
- B11 (a) $K_{\text{hp,max}} = 294.15/17 = 17.3$. (b) $P = 15/3.5 = 4.3 \text{ kW}$. (c) The resistance heater needs 15 kW : the heat pump uses 29% of the electricity
- B12 (a) $e_C = 1 - 278.15/300.15 = 0.073$ (7.3%). (b) $P_H \geq 10/0.0733 = 136 \text{ MW}$. (c) The real plant has a still smaller temperature difference across the heat exchangers, pumping power and irreversible heat flow
- B13 (a) Engine: $e_C = 0.50$, $W = 500 \text{ J}$, $Q_C = 500 \text{ J}$. Refrigerator: $K_C = 250/50 = 5.0$, so it removes $5.0 \times 500 = 2500 \text{ J}$. (b) The refrigerator rejects $2500 + 500 = 3000 \text{ J}$; with the engine's 500 J , the 300 K reservoir gains 3500 J . (c) $\Delta S = -\frac{1000}{600} - \frac{2500}{250} + \frac{3500}{300} = -1.667 - 10.0 + 11.667 = 0$: both machines are reversible, so the total entropy change is zero

Exam-style practice: answer key and brief solutions

Key: E1 B E2 E E3 A E4 B E5 D E6 D E7 C E8 D E9 C E10 C

E1 (B). $W = Q_h - Q_c = 280 \text{ J}$ per cycle (energy conservation over a cycle), so $e = W/Q_h =$

$280/800 = 0.35$. The ratio $Q_c/Q_h = 0.65$ is the fraction wasted.

E2 (E). $e_C = 1 - T_C/T_H = 1 - 300/500 = 0.40$ (temperatures in kelvin). The ratio $T_C/T_H = 0.60$ is the fraction rejected.

E3 (A). $W = eQ_h = (0.40)(2000) = 800$ J using the efficiency from the previous item. The remaining 1200 J is rejected to the cold reservoir.

E4 (B). I and II are statements of the second law. III is false: for an isolated system $\Delta S \geq 0$, with equality only for reversible processes. Answer: I and II only.

E5 (D). $K = Q_c/|W| = 400/100 = 4.0$. The ratio $Q_h/W = 5.0$ is the coefficient of performance of a heat pump, not of a refrigerator.

E6 (D). Energy conservation over a cycle: $Q_h = Q_c + |W| = 400 + 100 = 500$ J. The kitchen receives more heat than the refrigerator removes, which is why a fridge door left open warms a room.

E7 (C). $\Delta S = Q/T = mL_f/T = (0.500)(3.34 \times 10^5)/273.15 = 611$ J/K (positive: the liquid is more disordered). Use kelvin for T . The value 1.67×10^5 is the heat Q in joules, not an entropy.

E8 (D). Block: $\Delta S = mc \ln(T_f/T_i) = 195 \ln(300/360) = -35.6$ J/K. Lake: $Q/T = 11\,700/300 = +39.0$ J/K. Total = $+3.4$ J/K > 0 , as required for an irreversible process; the block alone decreases, but the lake gains more.

E9 (C). $e = 1 - r^{1-\gamma} = 1 - 8^{-0.40} = 1 - 0.435 = 0.565$. The ratio $8^{-0.40} = 0.435$ is the fraction of the heat input rejected. The result must be less than the Carnot efficiency for the same extreme temperatures.

E10 (C). The Carnot limit is $e_{\max} = 1 - 300/500 = 0.40$. I: $e = 300/1000 = 0.30 \leq 0.40$, possible. II: $e = 0.45 > 0.40$, impossible. III: $e = 300/800 = 0.375 \leq 0.40$, possible. Answer: I and III only.